University of Balamand

Department Of Computer and Electrical Engineering

Smart Migraine Monitoring Device

Lana Chaar, Nicolas Darwish and Rafic Srour

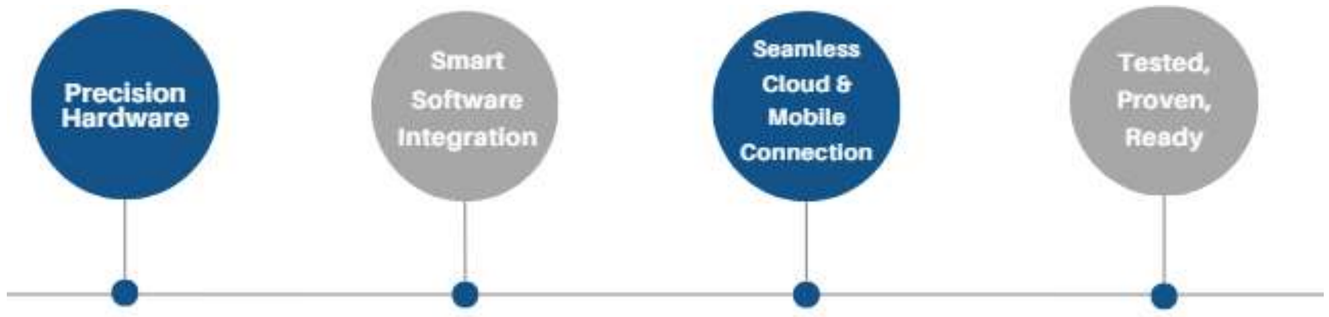
Introduction

What if a **migraine** didn't have to arrive without warning? What if a silent signal could tell you to slow down, breathe, and take control before the pain begins?

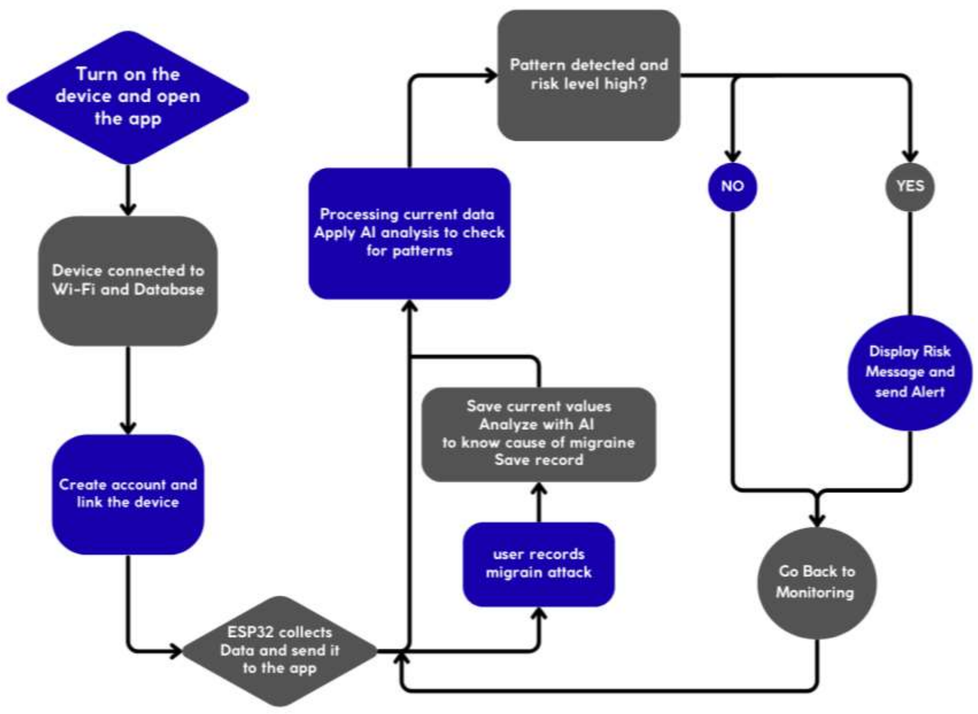
Migraines affect millions of people worldwide, often striking unexpectedly and disrupting daily life, work, and well-being. The **Smart Migraine Monitoring Device** reimagines migraine management by transforming invisible **physiological** and **environmental** changes into meaningful insights. Through continuous monitoring of **heart rate**, **temperature**, and **humidity**, the device listens to the body and its surroundings, **identifying patterns** that may signal an approaching migraine.

Designed as a **wearable**, **battery-powered** system with real-time cloud connectivity, the Smart Migraine Monitoring Device delivers timely, discreet alerts through haptic feedback and a mobile application. By turning data into **awareness** and **awareness into action**, the device empowers users to **anticipate migraines**, **reduce their impact**, and **regain confidence** in everyday life.

Design



General Block Diagram



Results

Continuous Migraine Monitoring:
The device continuously tracks physiological and environmental parameters associated with migraine episodes.

Early Warning & Prevention:
By detecting previously recorded patterns and changes in heart rate, temperature, and humidity, and taking into consideration user's profile information, the system helps warn the user and anticipate migraine risks before symptoms escalate.

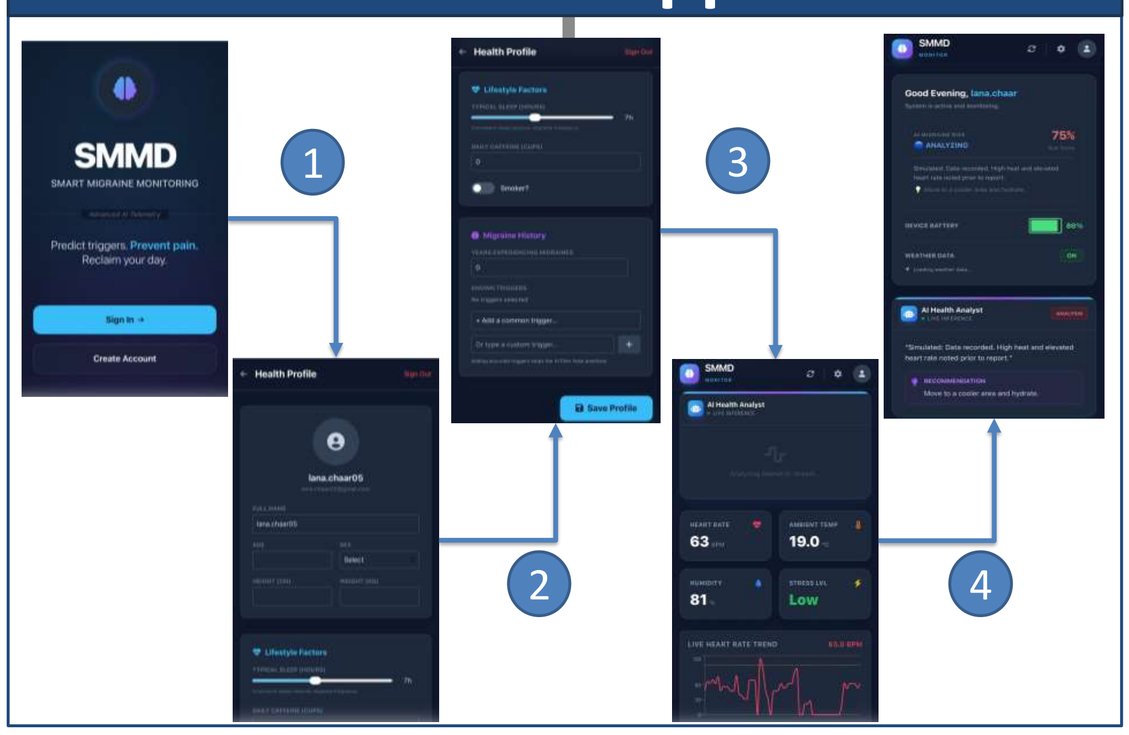
Real-Time Alerts:
Discreet haptic alerts notify users instantly, allowing timely action without disrupting daily activities.

Seamless Connectivity:
Wireless synchronization with the cloud and mobile application ensures real-time data access and monitoring.

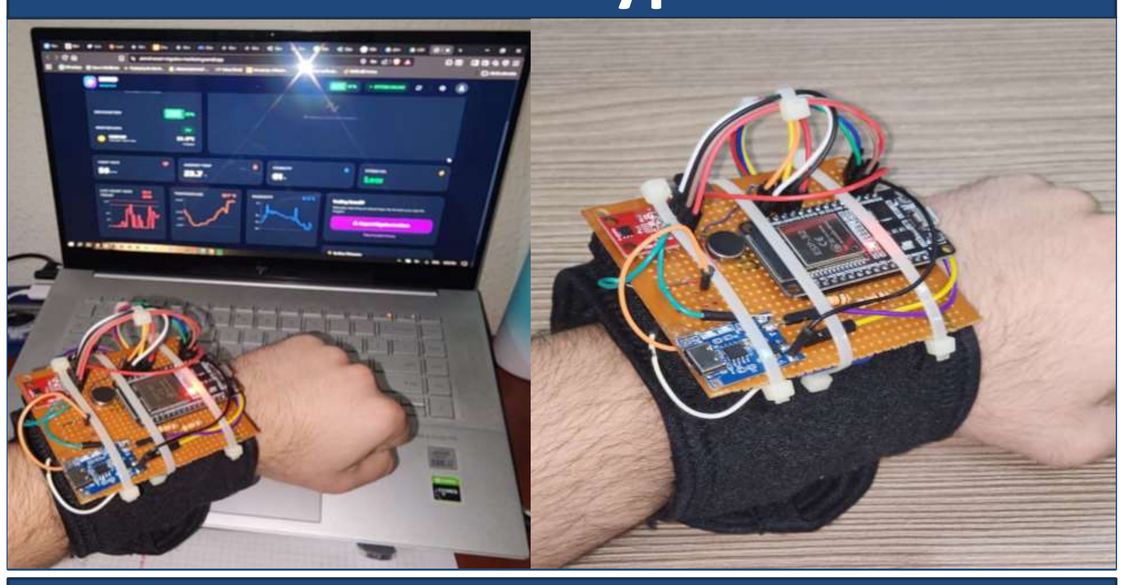
Safe & Wearable Design:
Battery-powered operation and tested hardware–software integration provide a reliable and user-friendly solution.

Future-Ready System:
The scalable architecture supports future AI-based migraine prediction and personalized health insights.

Mobile App



Prototype



Conclusion

A smart migraine monitoring device that brings **awareness**, **prevention**, and **control back to daily life**.

- Monitors key physiological and environmental migraine triggers
- Provides timely, discreet alerts through real-time connectivity
- Supports users in managing migraines with confidence and independence

References

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